Decoding Mental Health States from Social Media Text: A Multi-Class Classification Approach with Clinical Symptom-Alignment Validation

1. Problem Statement

Most machine learning models for detecting mental health conditions from social media report accuracy and F1-score but do not verify whether the model is actually learning clinically meaningful symptom language or is instead picking up spurious dataset artifacts (subreddit names, slang, posting patterns). This project builds a multi-class classifier that decodes mental health states — depression, anxiety, PTSD, stress, and baseline — from user-generated text, and then uses explainability methods to quantify how much of the model's decision-making aligns with clinically validated symptom indicators (PHQ-9, GAD-7, PCL-5) versus dataset noise.

2. Research Gap

Three separate clusters of prior work exist, but none combine all three elements needed for this contribution:

- Cluster A — Social media + multi-class + SHAP, no clinical grounding: existing papers classify multiple mental health categories from Reddit/Twitter with SHAP explainability, but SHAP output stays at the raw word/feature level and is never scored against a clinical symptom taxonomy.
- Cluster B — Clinical-scale + SHAP validation, no social media text: other papers validate SHAP outputs against PHQ-9/GAD-7 scores, but on structured clinical assessments, speech, or multimodal app data — not organic free-text posts.
- Cluster C — A 2024 arXiv survey ("Explainable AI for Mental Disorder Detection via Social Media") reviews the field and explicitly flags clinical validation of explainability outputs as an open, underdeveloped problem.

Contribution: this project applies the validation method from Cluster B to the classification setting in Cluster A — the first systematic scoring of whether a multi-class social-media mental-health classifier's SHAP explanations align with validated clinical symptom scales, rather than reporting accuracy alone.

3. Scope

- Classes: 4–5 categories — Depression, Anxiety, PTSD, Stress, Baseline/Control.
- Data modality: text-only, single platform (Reddit), single existing labeled dataset. No multimodal data, no cross-platform comparison, no primary data collection.
- Models: TF-IDF / BERT embeddings → XGBoost baseline, compared against a fine-tuned DistilBERT/RoBERTa model.

- Explainability: SHAP applied to the XGBoost model for feature-level interpretability.
- Core contribution: a symptom-alignment scoring framework mapping top SHAP features per class to PHQ-9 (depression), GAD-7 (anxiety), and PCL-5 (PTSD) item keywords.
- Explicitly out of scope: individual-level risk prediction, real-time deployment, or any diagnostic claim. The paper will state this as an ethical boundary.

4. Dataset

Primary dataset: Reddit Mental Health Dataset (RMHD) — publicly available, pre-labeled by subreddit (r/depression, r/Anxiety, r/PTSD, r/stress, r/CasualConversation as baseline). Widely used in prior literature, which keeps results comparable to existing benchmarks. No original user data will be scraped or self-labeled, avoiding the ethical issues associated with inferring diagnoses on unconsented individuals.

5. Ethical Considerations

- Only public, pre-anonymized, aggregate-level data will be used — no re-identification attempts.
- The system will not output any individual-level risk score or flag; it is a research/validation study, not a screening tool.
- The paper will include an explicit statement that the model is not diagnostic and is not intended for clinical or moderation deployment.

6. Timeline (6–8 weeks)

- Week 1: Dataset acquisition, cleaning, exploratory analysis.
- Week 2–3: Baseline model training (TF-IDF + XGBoost) and transformer fine-tuning (DistilBERT/RoBERTa).
- Week 4: SHAP explainability analysis on the trained classifier.
- Week 5: Build PHQ-9/GAD-7/PCL-5 symptom lexicon; develop alignment-scoring method.
- Week 6: Run alignment scoring, analyze spurious vs. clinically valid features.
- Week 7–8: Write-up, revisions, target journal formatting and submission.

7. Key Prior Work Reviewed

1. Trade-offs between ML and DL for mental illness detection on social media — Scientific Reports

2. Explainable ML for mental health prediction from social media behavior — Discover Mental Health, Springer

3. A Machine Learning Approach for Detection of Mental Health Conditions and Cyberbullying from Social Media — AAAI-26 Bridge / PMLR

4. Enhanced Suicidal Ideation Detection Using a CNN-BiLSTM Hybrid Model

5. Explainable AI for Mental Disorder Detection via Social Media: A Survey and Outlook

6. Explainable AI-driven depression detection from social media using NLP and black-box ML models — Frontiers in AI

7. Integrating explainable AI with clinical features to enhance ADHD diagnostic understanding (PHQ-9/GAD-7 + XGBoost + SHAP)

8. Predicting Mood Disorder Symptoms with Remotely Collected Videos Using an Interpretable Multimodal Dynamic Attention Fusion Network

9. Exploration of Perceptual Speech Features for Clinical Decision-Support in Mental Health Care

8. Target Venue Tier

Realistic targets given undergraduate authorship and timeline: Q3/Q4 journals or reputable open-access venues in computational health/NLP (e.g., JMIR Mental Health, Digital Health, IEEE JBHI-adjacent open-access outlets) — not top-tier venues, which is an appropriate and strategic scoping decision for a first paper.